Bacterial Genetics

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Escherichia coli chromosome (Nucleoid):
consists of a single circular DNA molecule composed of approximately 5×10^6 base pairs, that code for about 2000 proteins.

Bacterial cells are haploid; have a single chromosome.

- No nuclear membrane, no nucleolus, no mitotic spindle, no histones and no introns.

In haploid cells, any gene that has mutated—and therefore is not expressed—results in a cell that has lost that trait.

Prokaryote vs. Eukaryote Genetics

- **Prokaryotes**

  - Are haploid, and they contain a single circular chromosome

  - prokaryotes contain small circular DNA molecules called "plasmids".

- **Eukaryotes**

  - Are often diploid

  - Eukaryotes have linear chromosomes (usually more than 1

Bacterial Genetics

Other genetic elements

- Plasmids,
- Transposons, viruses (bacteriophages)

- Reproduction is asexual
- Poor ability to control environment

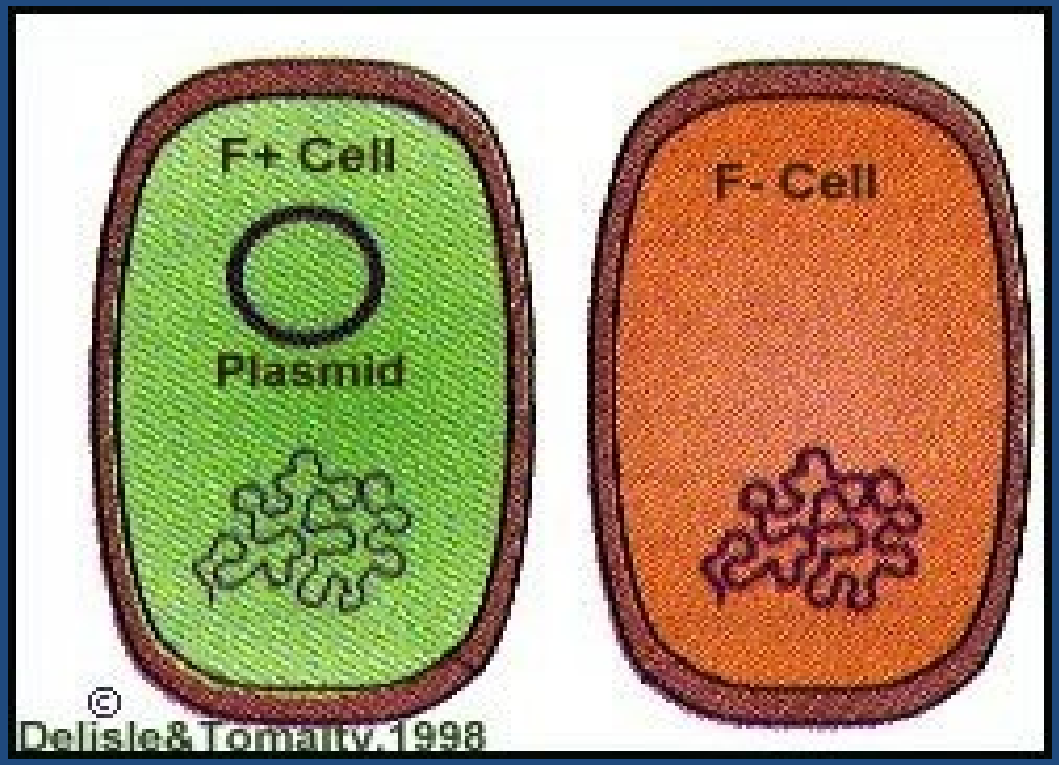
Plasmids:

Plasmids are extrachromosomal, double-stranded, circular DNA molecules that are capable of replicating independently of the bacterial chromosome. Plasmids occur in both gram-positive and gram-negative bacteria.

Types:

- **Transmissible plasmids (Conjugative plasmids)**: can be transferred from cell to cell by conjugation. They are large since they **code sex pilus** and enzymes required for genetic material transfer. They are usually present in a few (1–3) copies per cell.

- **Nontransmissible plasmids** are small since they do not contain the transfer genes; present in many (10–60) copies per cell.


F+ Cell
Plasmid
F- Cell
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Plasmids

- **Conjugative plasmids:** F factor (fertility factor) from *E. coli* is the most studied. F factor carries genes encoding the sex pilus for physical transfer of genetic material (horizontal transmission).

- **R factors:** multiple drug resistance (MDR plasmid)

- **Hfr - conjugative plasmid** integrated into the chromosome has the characteristic of high frequency of recombination, and passes this trait to any receptive bacteria with which the host cell conjugates

Plasmids genes serve for the following functions:

1- Antibiotic resistance, mediated by a variety of enzymes, such as the beta-lactamase of *S. aureus*, *Escherichia coli*, and *Klebsiella pneumoniae*.

2- Exotoxins, such as the enterotoxins of *E. coli*, anthrax toxin of *Bacillus anthracis*, and tetanus toxin of *Clostridium tetani*.

3- Pili (fimbriae !!), which mediate the adherence of bacteria to epithelial cells.

4- Resistance to ultraviolet light, which is mediated by DNA repair enzymes. (photolyase , DNA glycolase and others).

Other plasmid-encoded products :

Bacteriocins are toxic proteins produced by certain bacteria that are lethal for other bacteria. Two common mechanisms of action of bacteriocins are:

1- degradation of bacterial cell membranes by producing pores in the membrane.

2- degradation of bacterial DNA by DNAase.

Examples of bacteriocins produced by medically important bacteria are **colicins** made by *E. coli* and **pyocins** made by *Pseudomonas aeruginosa*.

- Bacteriocins may be useful in **treating infections caused by antibiotic-resistant bacteria.**

- A variety of degradative enzymes (coded by plasmids) that are produced by *Pseudomonas* and are capable of cleaning up environmental hazards such as oil spills and toxic chemical waste sites.

Bacterial Mutation

- Alteration of the information in a bacterial chromosome through a permanent change in the DNA.

Types of mutation:

1. Spontaneous
2. Induced

Spontaneous Mutation

- Heritable changes to the base sequence in the DNA that occur because of natural phenomena
- Errors made during DNA replication (*DNA polymerase fidelity and processivity*).

- May occur for every 10^{6} to 10^{10} replications

- The mutant may survive, multiply and emerge as a dominant

- *Neisseria gonorrhoeae:* widespread antibiotic mutations

Induced mutation

- **Chemical or physical agents: could cause mutations**

- **Ultraviolet light (UV): induces crosslinking of adjacent thymine (thymine dimers).**

- **Radiations.**

  - **Examples of chemical agents:**
    - **Nitrous acid:**
    - **Base analog: 5-bromouracil (Acyclovir: antiviral)**

Chemicals act in several different ways.

- Nitrous acid and alkylating agents:

They alter the existing base so that it forms a hydrogen bond with the wrong base

- 5-bromouracil (serves as base analogues):

They resemble normal bases, 5-bromouracil can be inserted in place of thymine. Then, 5-bromouracil that has less hydrogen-bonding fidelity than does thymine, will bind to guanine with greater frequency. This results in a transition from an A-T base pair to a G-C base pair.

- Used as antiviral drug (iododeoxyuridine).

- Benzpyrene, which is found in tobacco smoke, bind to the existing DNA bases and cause frameshift mutations.

Mutation caused by X-rays and ultraviolet light:

X-rays have high energy and can damage DNA in three ways:

(a) by breaking the covalent bonds that hold the ribose phosphate chain together.
(b) by producing free radicals that can attack the bases.
(c) by altering the electrons in the bases and thus changing their hydrogen bonding.

- Ultraviolet radiation, causes the cross-linking of the adjacent pyrimidine bases to form dimers (thymine dimer / Photolyase).

Viruses:

Certain viruses, such as the bacterial virus Mu (mutator bacteriophage), cause a high frequency of mutations (frameshift mutations).

Three types of mutations:

1- Base substitution: One base is inserted in place of another. Produced by errors due to DNA polymerase. Or produced by mutagen that changes hydrogen bonding. This mutation can be of two types: a- missense mutation (cause different amino acids to be inserted). B- nonsense mutation (Insertion of stop codon, then no protein is produced).

2- Frameshift mutation. This occurs when one or more base pairs are added or deleted, that results in the production of an inactive protein.

3- Transposons or insertion sequences are integrated into the DNA.

Conditional lethal mutations: used in vaccine development; using influenza virus as a vector.

Temperature-sensitive organisms can replicate at a relatively low, permissive temperature (e.g., 32°C) but cannot grow at a higher, restrictive temperature (e.g., 37°C). (mutation that altered conformation of essential proteins).

The virus cannot grow at 37°C and hence cannot infect the lungs, but it can grow at 32°C in the nose, where it can replicate and induce immunity.

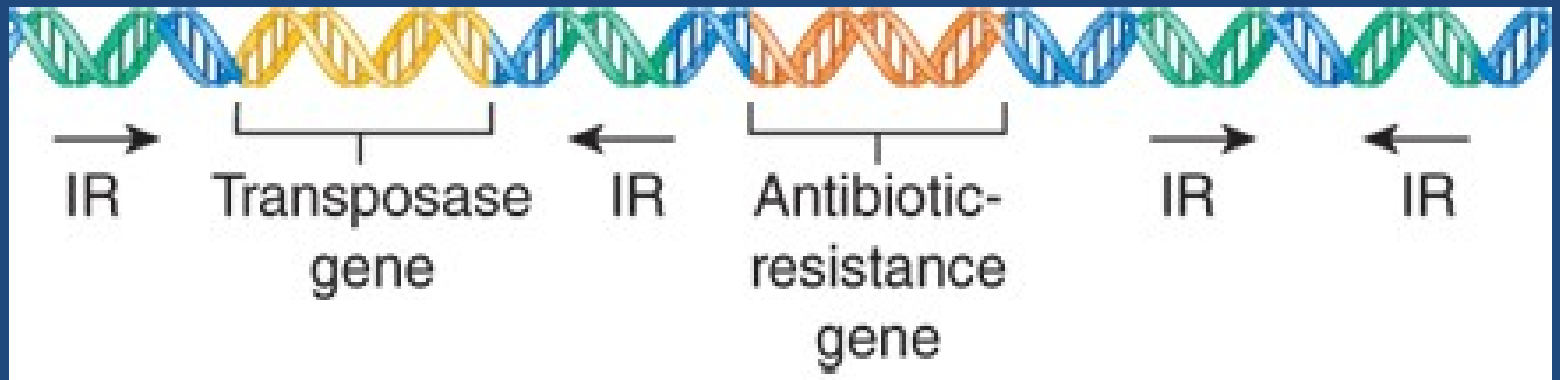
Transfer of DNA within bacterial cells

Transposons: are pieces of DNA that move readily from one site to another either within or between the DNAs of bacteria, plasmids, and bacteriophages.

- They are nicknamed "**jumping genes**".

Transposons can code for drug-resistant enzymes, toxins, or a variety of metabolic enzymes and can cause mutations or alter the expression of nearby genes.

Transposons typically have four identifiable domains. On each end is a short DNA sequence of inverted repeats, which are involved in the integration of the transposon into the recipient DNA. The second domain is the gene for the transposase, which is the enzyme that mediates the excision and integration processes. The third region is the gene for the repressor that regulates the synthesis of both the transposase and the protein encoded by the fourth domain, which, in many cases, is an enzyme mediating antibiotic resistance.



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Transposon genes: replicate as part of the bacterial chromosome

IR, inverted repeat (or IS: insertion sequence).

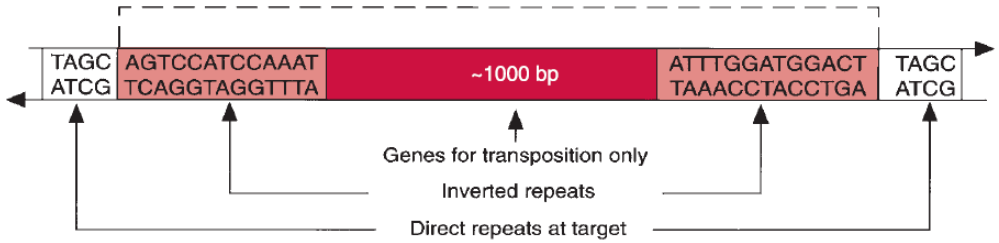
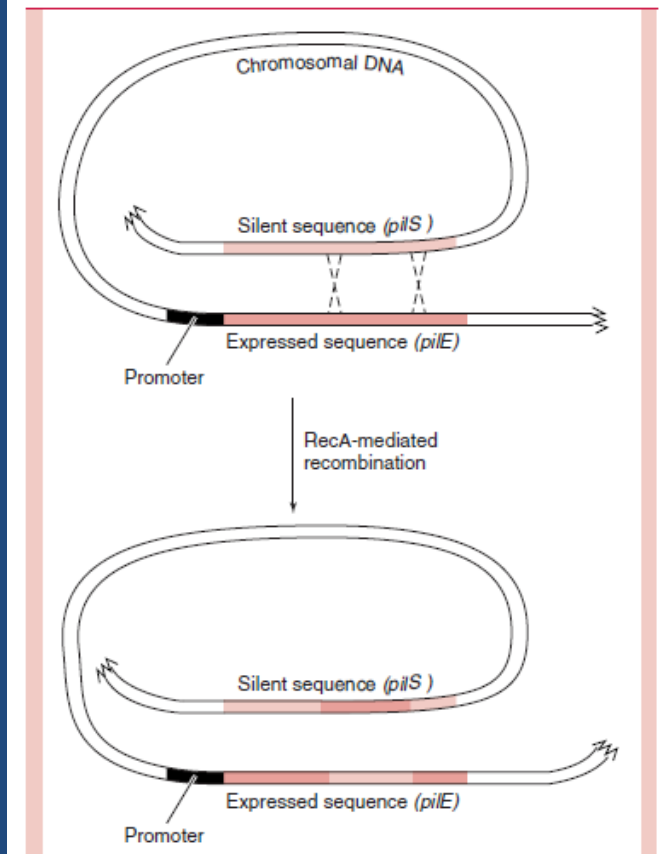


FIGURE 4–8

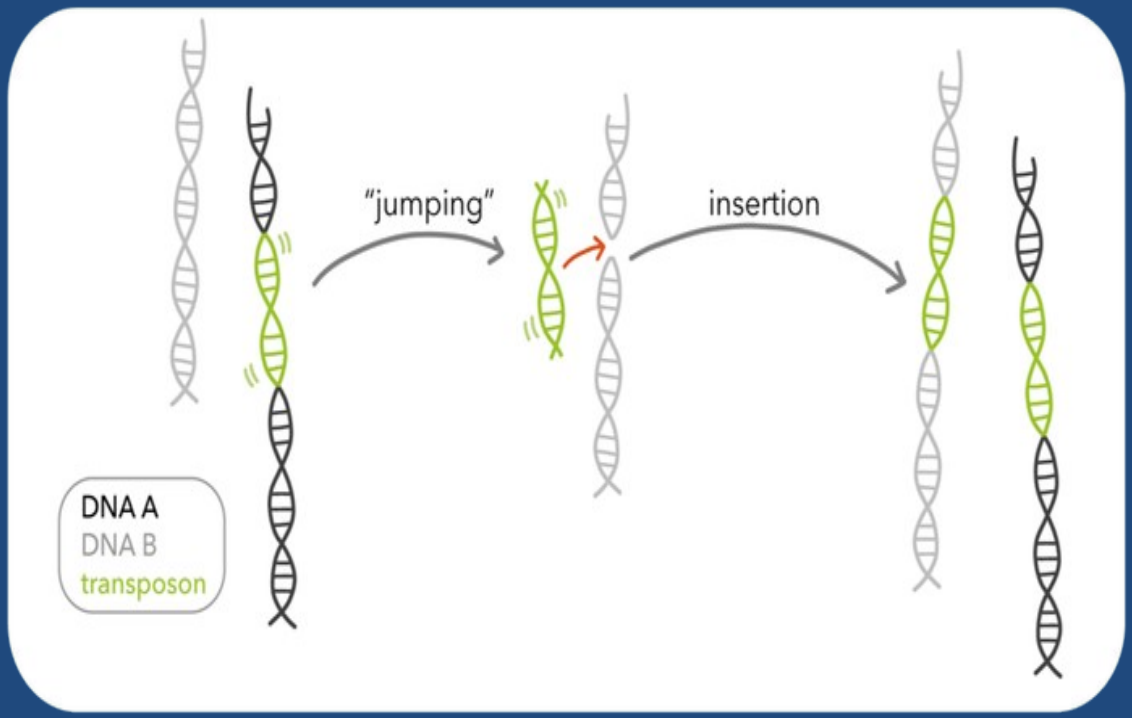
Structure of an insertion sequence (IS) element. The general features of bacterial IS elements are illustrated. As an example, IS2 has a total of 1327 bp, of which there are terminal inverted repeat sequences of 41 bp flanking the central region that encodes the one or two proteins required for transposition of IS2. A direct repeat of 5 bp was created at the site of insertion of the element. Approximately five IS2 elements are found in the chromosome of many strains of *Escherichia coli*.



TRANSFER OF DNA WITHIN BACTERIAL CELLS:

- Transposons transfer DNA from one site on the bacterial chromosome to another site or to a plasmid.

- The transfer of a transposon to a plasmid and the subsequent transfer of the plasmid to another bacterium by conjugation contributes significantly to the spread of antibiotic resistance.



Transposon insertions into the genome.
a transposon (green) is shown jumping from one piece of DNA to another.

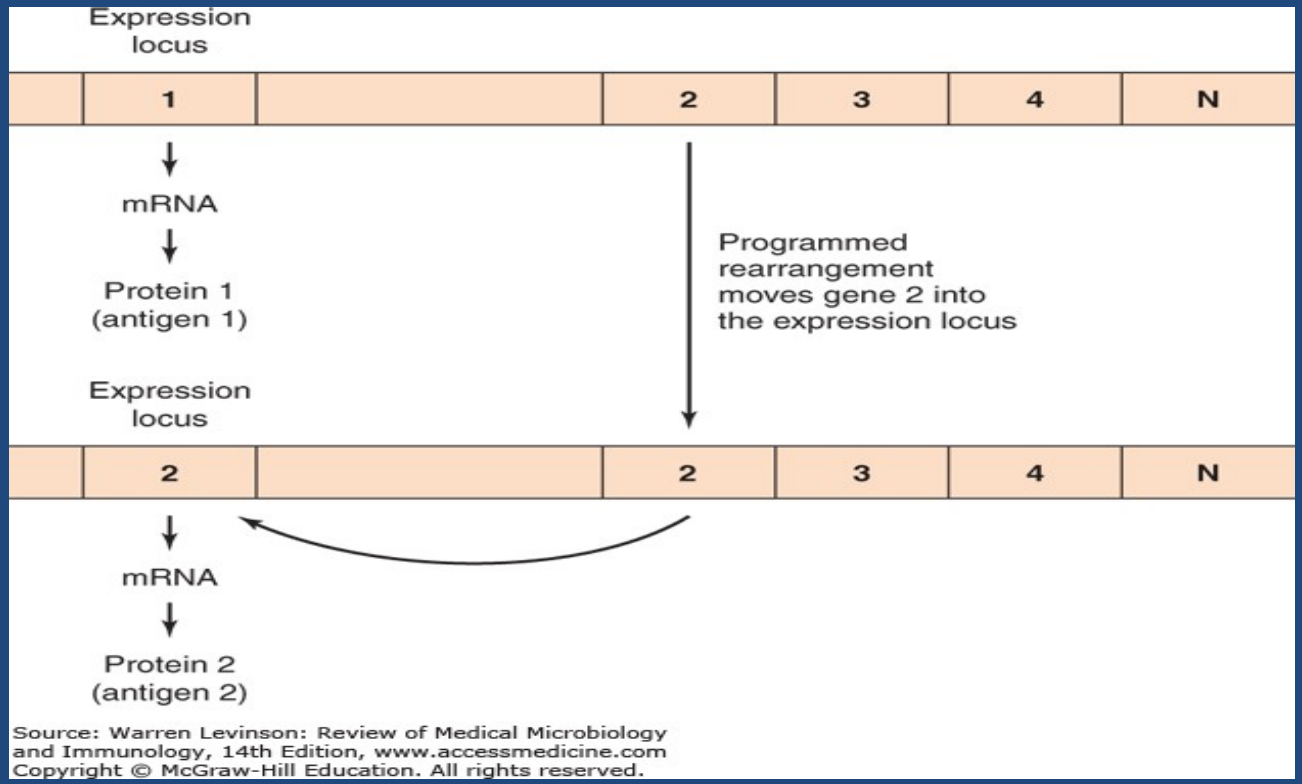
Transposons and programmed rearrangement

- **Programmed rearrangements**: Transfer of DNA within bacteria that responsible on many **antigenic changes** seen in *Neisseria gonorrhoeae* and *Borrelia recurrentis*.

- A programmed rearrangement consists of the movement of a gene **from a silent storage site** (no expression) **to an active site** where transcription and translation occur. The purpose is to **evade the immune response.**

Programmed rearrangements.

A copy of gene 2 is made and inserted into the expression locus. When the DNA of gene 2 is inserted, the DNA of gene 1 is excised and degraded.



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Transfer of DNA between Bacterial Cells

1. **Transformation** - Bacteria take up DNA from their environment and incorporate it into their genome (i.e. the Griffith experiment)

2. **Conjugation** - The direct transfer of DNA by bacteria usually via plasmids

3. **Transduction** - Movement of DNA between bacteria by viruses

TRANSFER OF DNA BETWEEN BACTERIAL CELLS

This may affect:

1- **Antibiotic resistance** genes are spread from one bacterium to another primarily by conjugation.

2- Several important **exotoxins** are encoded by bacteriophage genes and are transferred by transduction.

1. Conjugation

Transfer of genetic material from donor to recipient cell. The mating process is controlled by an F (fertility) plasmid (F factor), which carries the genes required for conjugation.

One of the most important proteins is pilin, which forms the sex pilus (conjugation tube).

Mating begins when the pilus of the donor bacterium carrying the F factor (F^+) attaches to a receptor on the surface of the recipient bacterium (F^-).

After an enzymatic cleavage of the F factor DNA, one strand is transferred across the conjugal bridge (sex pilus) into the recipient cell. The process is completed by synthesis of the complementary strand to form a double-stranded F factor plasmid in both the donor and recipient cells.
- in this instance only the F factor, and not the bacterial chromosome, has been transferred.

Resistance plasmids (R plasmids): can carry one or more genes for a variety of enzymes that can degrade antibiotics and modify membrane transport systems.

For example, R plasmids encode the **β-lactamases** of *Staphylococcus aureus*, *E. coli*, and *Klebsiella pneumoniae*.

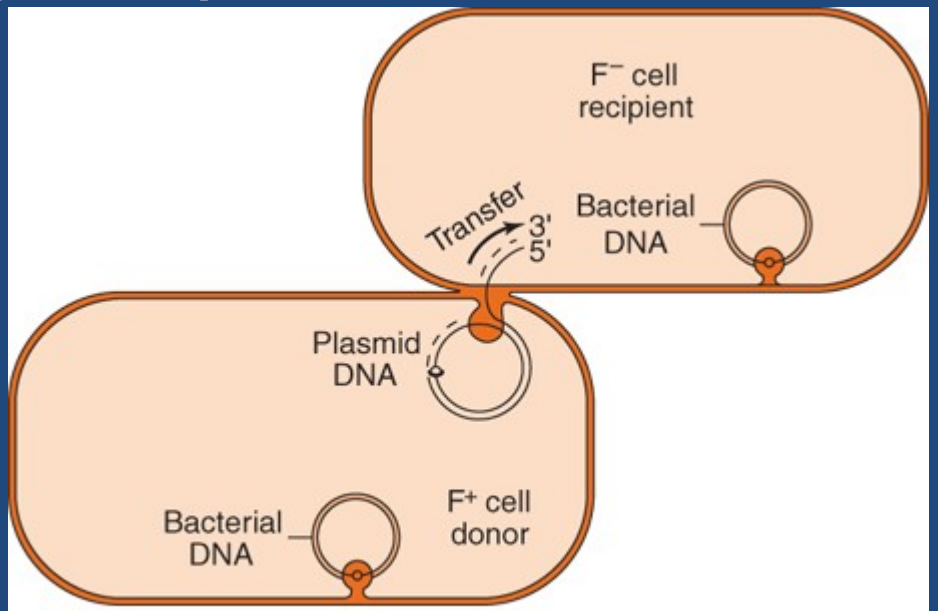
R plasmids are transferred **by conjugation** not only to cells of the same species, **but also to other species.**

- Conjugation:

- An F plasmid is being transferred from an F^+ donor bacterium to an F^- recipient.

- The new plasmid in the recipient bacterium is composed of one parental strand (solid line) and one newly synthesized s trand (dashed line).

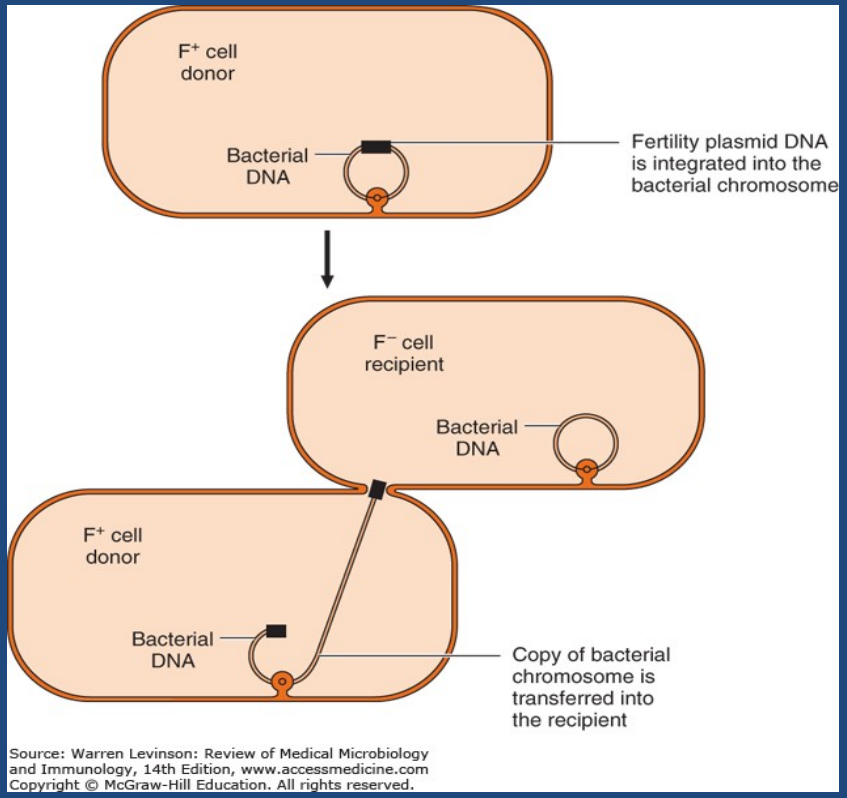
- At the end of DNA synthesis, both the donor and the recipient contain a complete copy of the plasmid DNA.



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High-frequency recombination:

Top: A fertility (F) plasmid has integrated into the bacterial chromosome.



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Bottom: The F plasmid mediates the transfer of the bacterial chromosome of the donor into the recipient bacteria.

2. Transduction: is the transfer of cell DNA by means of a bacterial virus (bacteriophage).

- The virus can take a piece of bacterial DNA; incorporate it into the virus particle and then transfer it into new recipient cell at the next time of infection.

- Within the recipient cell, the phage DNA can integrate into the cell DNA and the cell can acquire a new trait—a process called lysogenic conversion .

- This process can change a nonpathogenic organism into a pathogenic one.

- Diphtheria toxin, botulinum toxin, cholera toxin, and erythrogenic toxin (*Streptococcus pyogenes*) are encoded by bacteriophages and can be transferred by transduction.

Transduction.

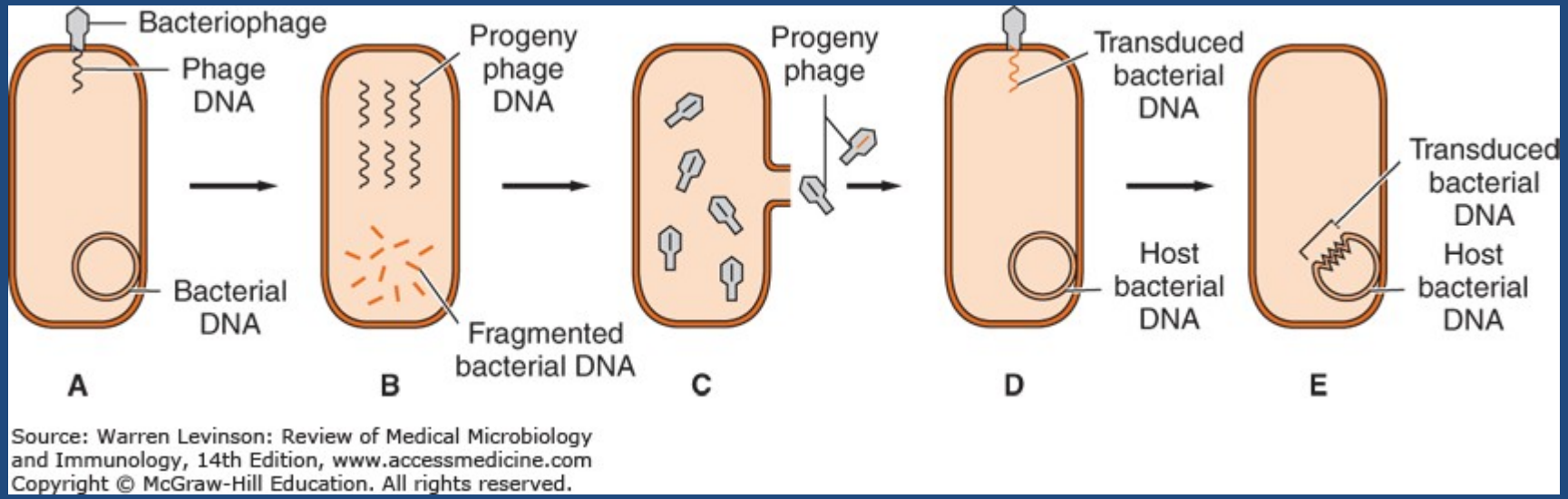
A: A bacteriophage infects a bacterium, and phage DNA enters the cell.

B: The phage DNA replicates, and the bacterial DNA fragments.

C: The progeny phages assemble and are released; (a few contain bacterial DNA).

D: Another bacterium is infected by a phage-containing bacterial DNA.

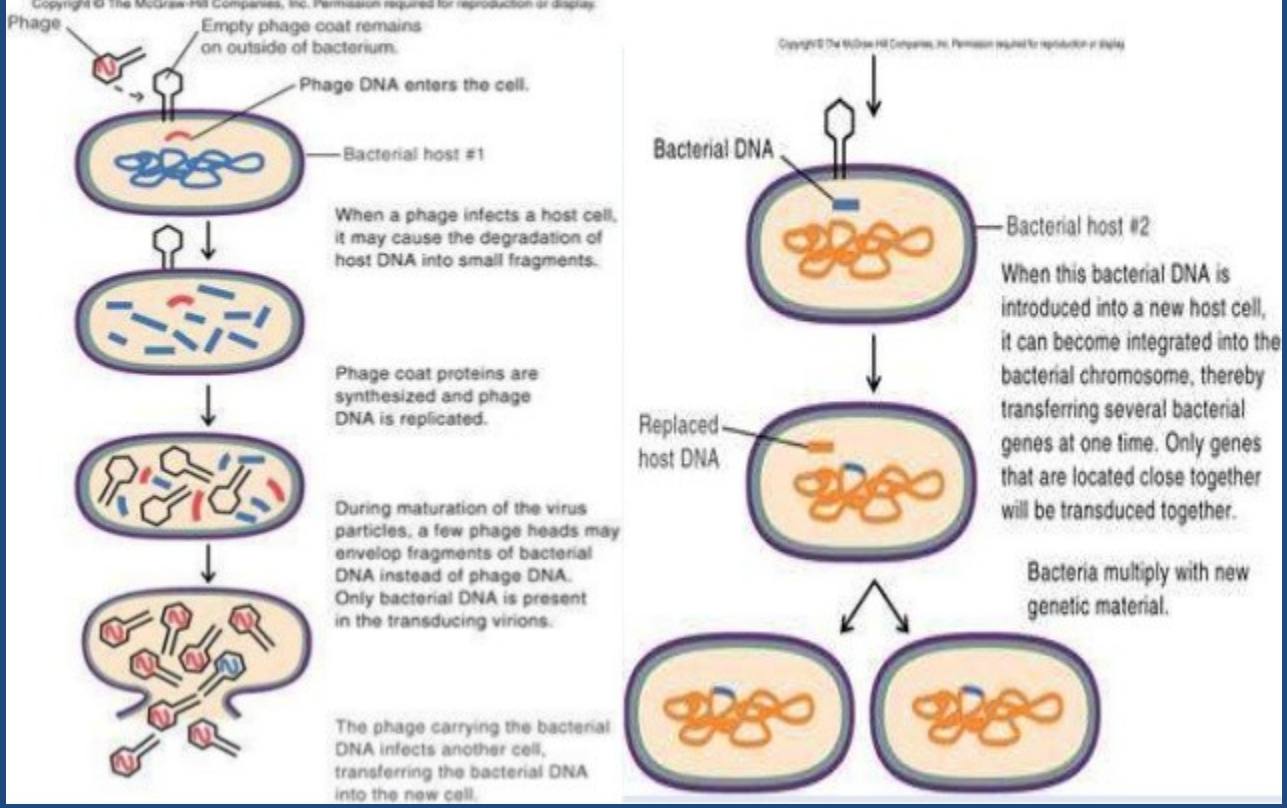
E: The transduced bacterial DNA integrates into host DNA, and the host acquires a new trait.



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GENERALIZED TRANSDUCTION

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3. Transformation: **is the transfer of DNA from one cell to another. This occurs by:**

1- In nature, DNA of dying bacteria may be taken up by recipient cells. Example: *Neisseria* and *Haemophilus*, synthesize receptors on the cell surface that bind and uptake of DNA from the environment.

2- In the laboratory: Done by extracting DNA from one type of bacteria and introduce it into genetically different bacteria. Also, part of the DNA can be transferred using plasmid.
- Transfection: if DNA was injected into the cell or the nucleus of an eukaryotic cell)

Transformation experiment: In 1944, it was shown that DNA extracted from encapsulated smooth pneumococci could transform nonencapsulated rough pneumococci into encapsulated smooth organisms. This demonstration that the transforming principle was DNA.

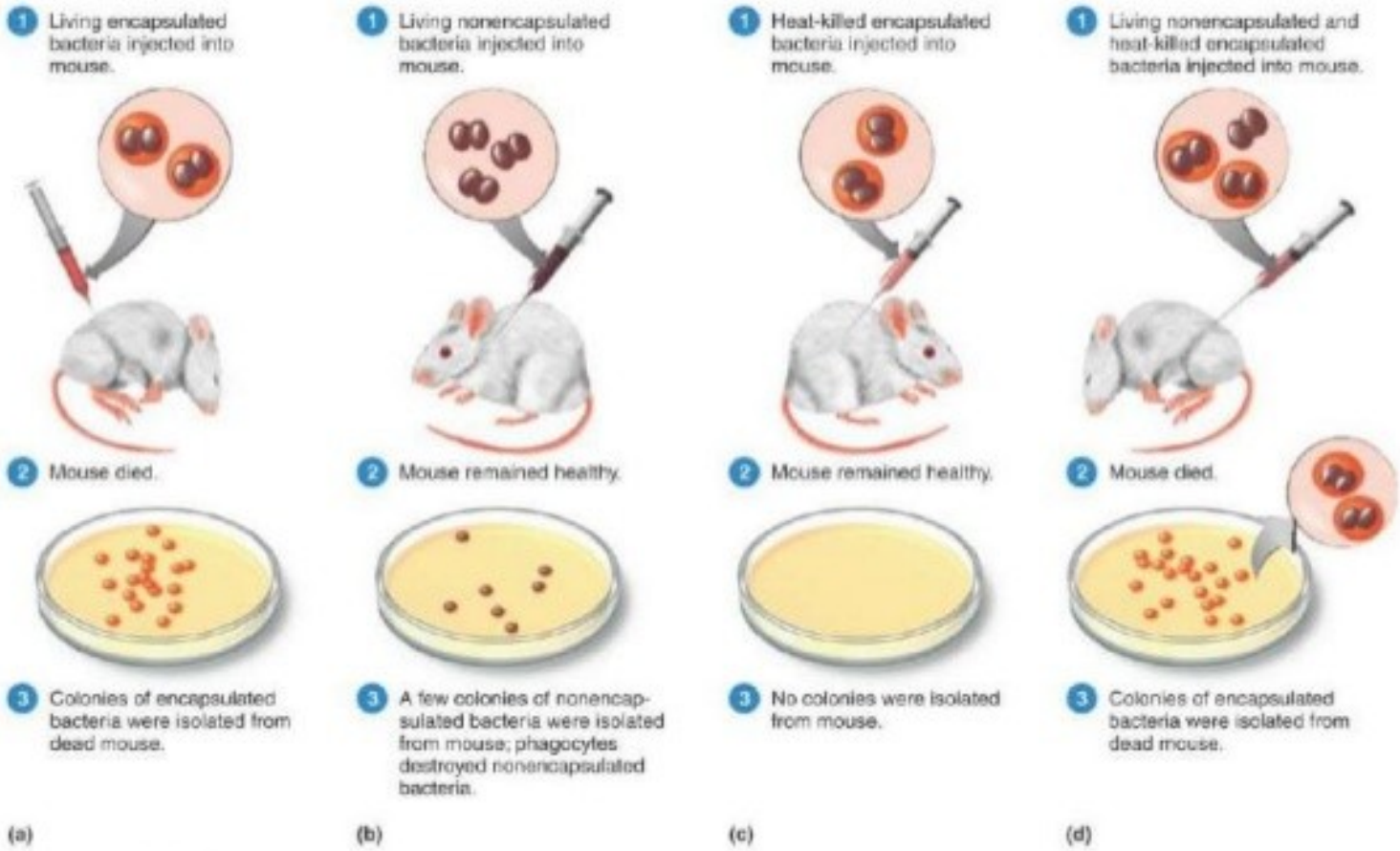


Figure 8.24 Griffith's experiment demonstrating genetic transformation. (**a**) Living encapsulated bacteria caused disease and death when injected into a mouse. (**b**) Living nonencapsulated bacteria are readily destroyed by the phagocytic defenses of the host, so the mouse remained healthy after injection. (**c**) After being killed by heat, encapsulated bacteria lost the ability to cause disease. (**d**) However, the combination of living nonencapsulated bacteria and heat-killed encapsulated bacteria (neither of which alone cause disease) did cause disease. Somehow, the live nonencapsulated bacteria were transformed by the dead encapsulated bacteria so that they acquired the ability to form a capsule and therefore cause disease. Subsequent experiments proved the transforming factor to be DNA.

Comparison of Conjugation, Transduction, and Transformation

Transfer Procedure	Process	Type of Cells Involved	Nature of DNA Transferred
Conjugation	DNA transferred from one bacterium to another	Prokaryotic	Chromosomal or plasmid
Transduction	DNA transferred by a virus from one cell to another	Prokaryotic	Any gene in generalized transduction
Transformation	Purified DNA taken up by a cell	Prokaryotic or eukaryotic	Any DNA

RECOMBINATION:

Once the DNA is transferred from the donor to the recipient cell, it can integrate into the host cell chromosome by recombination. There are two types of recombination:

- Homologous recombination, in which two pieces of DNA that have extensive homologous regions pair up and exchange pieces by the processes of breakage and reunion.

- Nonhomologous recombination, in which little, if any, homology is necessary.

Different genetic loci govern these two types, and different enzymes are involved. (endonucleases and ligases).